

SOIL DOCTOR : EMBEDDED AI-BASED SOIL ANALYSIS DEVICE FOR CROP AND FERTILIZER RECOMMENDATION

A MINI PROJECT REPORT

Submitted by

Gopika Suresh : KKE23EC028
Megha C : KKE23EC036
Midhunrag N V : KKE23EC037
Shreya S N : KKE23EC062

to

the APJ Abdul Kalam Technological University
in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

in

Electronics and Communication Engineering



**Department of Applied Electronics & Instrumentation
Engineering**

Government Engineering College - Kozhikode
Kannur Road, Westhill
Kozhikode, Kerala 673005

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DECLARATION

We undersigned hereby declare that the project report “**SOIL DOCTOR : EMBEDDED AI-BASED SOIL ANALYSIS DEVICE FOR CROP AND FERTILIZER RECOMMENDATION**”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Dr. JESY P.** This submission represents our ideas in our own words and wherever ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Date: 6 April 2026

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DEPARTMENT OF APPLIED ELECTRONICS AND
INSTRUMENTATION ENGINEERING



CERTIFICATE

*This is to certify that the project entitled “**SOIL DOCTOR : EMBEDDED AI-BASED SOIL ANALYSIS DEVICE FOR CROP AND FERTILIZER RECOMMENDATION**” submitted by **GOPIKA SURESH (KKE23EC028)**, **MEGHA C (KKE23EC036)**, **MIDHUNRAG N V (KKE23EC037)**, and **SHREYA S N (KKE23EC062)** to the APJ Abdul Kalam Technological University, Kerala in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in **Electronics and Communication Engineering** is a bonafide record of the project work carried out by them under my supervision and guidance. This report in any form has not been submitted to any other University or Institute for any purpose.*

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Project Guide

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Dr. Baburaj M

Head of the Department

ABSTRACT

Agriculture plays a vital role in ensuring food security and sustainable development. Efficient soil management is essential for improving crop productivity and optimizing fertilizer usage. This project presents an Embedded AI-Based Soil Analysis Device designed to perform real-time soil monitoring and provide intelligent recommendations for crop selection and fertilizer application. The system utilizes a multi-parameter soil sensor to measure key soil characteristics such as moisture, temperature, pH level, electrical conductivity, and nutrient content including nitrogen, phosphorus, and potassium. The collected data is processed by an ESP32 microcontroller, which performs data acquisition, signal processing, and embedded AI-based analysis. A lightweight decision-making model is implemented to evaluate soil conditions and generate appropriate recommendations. The proposed device is portable, cost-effective, and capable of on-field operation, eliminating the need for traditional laboratory-based soil testing. The results are displayed through a user-friendly interface, providing farmers with real-time information about soil health, suitable crops, and required fertilizer adjustments. The integration of embedded systems and artificial intelligence enhances the accuracy and efficiency of soil analysis, contributing to precision agriculture and sustainable farming practices.

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LIST OF ABBREVIATIONS

AI	– Artificial Intelligence
ANN	– Artificial Neural Network
API	– Application Programming Interface
CRC	– Cyclic Redundancy Check
EC	– Electrical Conductivity
ESP32	– Espressif 32-bit Microcontroller
GPIO	– General Purpose Input Output
I2C	– Inter-Integrated Circuit
IDE	– Integrated Development Environment
IoT	– Internet of Things
KNN	– K-Nearest Neighbour
LCD	– Liquid Crystal Display
LIME	– Local Interpretable Model-Agnostic Explanations
LPWAN	– Low Power Wide Area Network
MAX485	– RS485 to TTL Converter Module
MCU	– Microcontroller Unit
ML	– Machine Learning
MLR	– Multiple Linear Regression
Modbus-RTU	– Remote Terminal Unit Protocol
NPK	– Nitrogen, Phosphorus, Potassium
PCB	– Printed Circuit Board
RF	– Random Forest
RS485	– Recommended Standard 485 (Serial Communication)
SDG	– Sustainable Development Goals
SHAP	– SHapley Additive exPlanations
SVM	– Support Vector Machine
TTL	– Transistor-Transistor Logic
TNAU	– Tamil Nadu Agricultural University
UI	– User Interface
XAI	– Explainable Artificial Intelligence

Chapter 1

INTRODUCTION

Agriculture plays a crucial role in ensuring food security and supporting economic development, and soil health is one of the most important factors influencing crop productivity. Among various soil parameters, the primary macronutrients—Nitrogen (N), Phosphorus (P), and Potassium (K)—have a direct impact on plant growth, yield, and overall soil fertility. Imbalanced nutrient levels can lead to reduced crop performance, excessive or insufficient fertilizer usage, environmental degradation, and long-term soil damage. Traditional soil testing methods involve manual sample collection and laboratory analysis, which, although accurate, are time-consuming, costly, and unsuitable for frequent field-level monitoring. Consequently, farmers often rely on experience-based fertilizer practices rather than real-time soil data.

Recent advancements in embedded systems and intelligent data processing techniques have enabled the development of smart agricultural solutions. Microcontroller platforms such as the ESP32 provide efficient processing capability, low power consumption, and suitability for portable field applications. By integrating multi-parameter soil sensors with decision-based algorithms, automated analysis of soil nutrient data and generation of appropriate crop and fertilizer recommendations can be achieved.

This project focuses on the design and implementation of an Embedded AI-Based Soil Analysis Device that measures soil NPK values and provides fertilizer recommendations using a Decision Tree-based inference model, thereby promoting precision agriculture, improving fertilizer efficiency, and supporting sustainable soil management practices. Based on the analyzed data, the system provides suitable crop suggestions and fertilizer recommendations, which are displayed through an LCD interface. The proposed solution is portable, cost-effective, and capable of delivering real-time insights, thereby assisting farmers in making informed agricultural decisions.

Overall, this project aims to bridge the gap between traditional soil testing practices and modern smart farming techniques by providing an efficient, automated, and user-friendly soil monitoring system that enhances agricultural productivity and promotes sustainable farming practices.

1.1 MOTIVATION

The motivation behind this project lies in transforming agricultural practices into data-driven systems that maximize productivity and environmental sustainability through embedded intelligence. In many farming communities, traditional laboratory-based soil testing creates significant bottlenecks due to high costs, time-consuming processes, and the impracticality of frequent monitoring. For example, a farmer might face an “Information Lag” of several weeks for a lab report, forcing them to rely on experience-based guesswork during critical planting windows. In rural areas, traditional assumptions are often more vulnerable to yield loss and soil degradation, as they rely on patrons’ adherence to conventional wisdom without scientific oversight.

In many regions, soil health serves as a crucial resource, but its potential is often limited by traditional laboratory systems and fixed testing cycles. Farmers and agricultural staff may need to assess nutrient levels immediately before sowing or after harvest, but waiting for laboratory results can disrupt their schedules and reduce crop quality. In addition, manual library-style processes for estimating fertilizer needs make it challenging to maintain nutrient accuracy, increasing the risk of chemical runoff and long-term land damage. The need for a more efficient, accessible, and real-time system becomes apparent when these limitations hinder a farmer’s learning opportunities for sustainable growth.

Automation through Embedded AI can alleviate these problems by enabling on-site, self-service diagnostics that minimize wait times, reduce human error, and improve soil security. With the “Soil Doctor” system, soil monitoring can be performed autonomously, allowing users to obtain NPK, pH, and moisture insights at any time. This increased accessibility not only benefits smallholder farmers by reducing costs but also empowers them to focus on precision agriculture rather than routine administration of assumptions. Through this project, we aim to demonstrate how embedded automation can elevate the functionality of soil management, making it more adaptive to users’ needs while enhancing the reliability and sustainability of agricultural resources.

Chapter 2

LITERATURE SURVEY

Several relevant studies and research papers were identified during the literature survey for the development of the Soil Doctor embedded AI system. The research highlights a significant shift from traditional lab-based testing toward real-time, automated, and intelligent agricultural monitoring.

According to Khaliq et al. (2025) in [1], the authors comprehensively explore an AI-driven smart agriculture framework that integrates 7-in-1 multifunctional sensors to collect real-time data on soil moisture, temperature, pH, and macronutrient (NPK) levels. The study emphasizes moving beyond traditional “fixed rules” or human intuition by utilizing advanced models like SwiFT for crop classification and TabNet for fertilizer recommendations. A critical innovation in this work is the application of Explainable AI (XAI) techniques, such as LIME and SHAP, to ensure that the AI’s reasoning is transparent and interpretable for the end-user.

According to Madhuri et al. (2022) in [2], the authors evaluate several supervised learning algorithms—including ANN, SVM, and KNN—for predicting crop yields based on soil parameters. The research specifically identifies the Decision Tree algorithm as the most effective method, achieving a 95% accuracy rate. The study highlights that Decision Trees are particularly suitable for agricultural applications because they provide clear, rule-based logic that is significantly easier to interpret and implement on resource-constrained embedded hardware compared to more complex models.

Furthermore, Madhumathi et al. (2023) in [3] explore the efficiency of various regression methods specifically for soil nutrient prediction. Their research reveals that the BORUTA regression model demonstrated the highest efficiency among tested models, achieving an accuracy rate of 91%. This study underscores the potential of data-driven approaches to accurately forecast soil health indicators, helping farmers optimize their resource usage and promote long-term sustainability.

Furthermore, in the Tamil Nadu Agricultural University (TNAU) Agritech Portal (2024) [4], standardized regional crop requirements are provided as a foundational database. This resource defines the optimal nutrient thresholds for six primary soil health indicators: Nitrogen (N), Phosphorus (P), Potassium (K), pH, Electrical Conductivity (EC), and Temperature. These benchmarks serve as the practical guidelines for maintaining soil health and are used to perform the gap analysis required for accurate fertilizer recommendations.

In Chowdam et al. (2025) [5], an IoT-enabled smart monitoring system is proposed that utilizes the MAX485 module to ensure reliable data transmission between sensors and microcontrollers in harsh field conditions. The system employs the Random Forest algorithm, achieving 90% accuracy in crop prediction. The authors highlight the necessity of closing the gap between technology breakthroughs and real-world agriculture by providing scalable and cost-effective solutions for on-site data gathering.

According to Sushma et al. (2024) in [6], the integration of IoT sensors and cloud computing enables continuous soil health monitoring, offering a promising solution to the “Information Lag” caused by delayed laboratory testing. Their proposed system utilizes an ESP32 platform to acquire data and transmit it wirelessly to cloud dashboards for analysis. By identifying and understanding these obstacles, automated systems can foster better knowledge dissemination and improve overall agricultural productivity.

Relevance to Proposed Work: The Soil Doctor project adopts the industrial 7-in-1 sensor and ESP32 platform described in Khaliq et al. to achieve high-precision data acquisition [1]. The selection of the Decision Tree Classifier for crop recommendation is directly supported by its 95% accuracy and explainable logic demonstrated by Madhuri et al [2]. While existing literature often focuses on cloud-dependent systems, this project innovates by extracting Decision Tree rules and implementing them as fully offline, edge-based logic on the ESP32, ensuring functionality in remote areas without internet access. Finally, the TNAU benchmarks are incorporated to ensure that fertilizer advice remains accurate and region-specific [4].

Chapter 3

OBJECTIVES

To address the current challenges in traditional agricultural practices, we identified the need to design and implement an Embedded AI-Based Soil Analysis Device. The objectives of the system are listed below:

- To enhance the agricultural experience for farmers: The system aims to provide convenient, real-time soil health assessments by simultaneously measuring the ‘Big Three’ macronutrients (NPK) along with pH, EC, moisture, and temperature through an integrated 7-in-1 sensor. This provides farmers with instant, laboratory-grade insights directly in the field.
- To reduce the information lag and eliminate guesswork: The reliance on traditional, time-consuming laboratory reports can be minimized by introducing an embedded AI recommendation engine. The system aims to help farmers instantly identify the most suitable crops and determine precise fertilizer requirements without the need for manual sampling or waiting weeks for results.
- To promote sustainable soil management and offline reliability: The project aims to prevent soil degradation and environmental damage by providing 95% accurate fertilizer recommendations, ensuring farmers avoid excessive chemical usage. It also aims to ensure proper functioning in remote rural areas by utilizing edge-based inference on the ESP32, removing any dependency on unreliable internet connectivity.
- To implement the system with minimum cost: Cost-effectiveness is a key consideration for widespread adoption by smallholder farmers. This objective emphasizes the need to develop a budget-friendly, portable solution using an affordable ESP32-based architecture that serves as a handheld alternative to expensive laboratory testing.

These objectives collectively aim to create an efficient, reliable, and cost-effective solution for managing agricultural activities conveniently, thereby enhancing the productivity and sustainability for the end users.

Chapter 4

PROPOSED METHOD

We have devised a solution to address the issues faced by conventional agricultural soil management practices. The details of the proposed method are as follows:

4.1 PROBLEM FORMULATION

Agricultural productivity is fundamentally dependent on maintaining balanced soil nutrient levels, particularly the primary macronutrients: Nitrogen (N), Phosphorus (P), and Potassium (K). However, traditional soil management often relies on manual observation and generalized practices that lack the precision needed for modern sustainable farming. These inefficiencies lead to a critical “Information Lag,” where the time between soil sampling and receiving laboratory results can take several weeks. This delay prevents timely intervention, forcing farmers to rely on experience-based guesswork for fertilizer application.

Such reliance on assumptions is highly prone to error, leading to improper fertilizer usage that results in soil degradation, environmental damage (such as chemical runoff), and reduced crop yields. Furthermore, existing high-tech solutions often suffer from connectivity dependency, requiring a stable internet connection that is frequently unavailable in remote rural areas. Consequently, small-scale farmers are burdened by high costs and a lack of actionable data at the field’s edge. Therefore, there is a critical need for a portable, standalone, and automated soil monitoring system that ensures real-time nutrient tracking, minimizes chemical waste, and optimizes overall agricultural management through offline intelligence.

4.2 EXISTING SYSTEMS

Existing systems for soil analysis primarily revolve around manual laboratory-based testing and subjective traditional knowledge. Laboratory methods involve the physical collection of soil samples which are then transported to a centralized facility for chemical analysis. While accurate, this process is time-consuming, expensive, and

impractical for the frequent monitoring required throughout a crop’s growth cycle.

Alternatively, many farmers utilize experience-based application, where fertilizer needs are estimated based on previous harvests or visual plant symptoms. This method is highly subjective and often fails to reflect the actual, dynamic nutrient profile of the soil, leading to significant imbalances. Even within current “smart” agricultural research, many systems focus on isolated components (such as only sensing moisture or NPK) or require complex cloud-based architectures that fail when internet access is lost. There is a clear need for a more advanced, integrated, and edge-based solution that provides convenient, on-site diagnostics and laboratory-grade recommendations without the drawbacks of manual testing or connectivity requirements.

4.3 PROPOSED SOLUTION

The proposed system aims to develop an Embedded AI-Based Soil Analysis Device capable of performing real-time soil monitoring and providing intelligent recommendations for crop selection and fertilizer usage. The system integrates sensing technology, embedded processing, and artificial intelligence to deliver a compact, portable, and cost-effective solution for modern agriculture.

Traditional soil testing methods rely on laboratory analysis, which is time-consuming, expensive, and unsuitable for frequent monitoring. To address these limitations, the proposed method focuses on an on-field, real-time analysis system that uses sensors to measure soil parameters and embedded AI algorithms to interpret the data instantly.

The system is built around a microcontroller-based architecture using the ESP32, which serves as the core processing unit. Various sensors are interfaced with the microcontroller to collect soil data, which is then analyzed using an embedded inference model. The system provides outputs in the form of crop recommendations and fertilizer suggestions, enabling farmers to make informed decisions directly in the field.

Chapter 5

SYSTEM CONCEPT

The system concept is centered on integrating real-time sensing, embedded processing, and intelligent decision-making into a single, autonomous portable device. The working of the system involves a multi-stage pipeline, beginning with the capture of seven critical soil characteristics: Nitrogen (N), Phosphorus (P), Potassium (K), pH, Electrical Conductivity (EC), moisture, and temperature. These parameters are essential indicators of soil health and directly influence agricultural productivity.

By utilizing an ESP32 microcontroller for local processing, the system eliminates the “Information Lag” of traditional laboratory testing. All processing is carried out at the edge, ensuring the device remains fully functional in remote areas without internet or cloud dependency. The core innovation is the transformation of raw sensor data into actionable intelligence, providing farmers with instant, lab-quality insights directly in the field.

5.1 OVERALL SYSTEM ARCHITECTURE

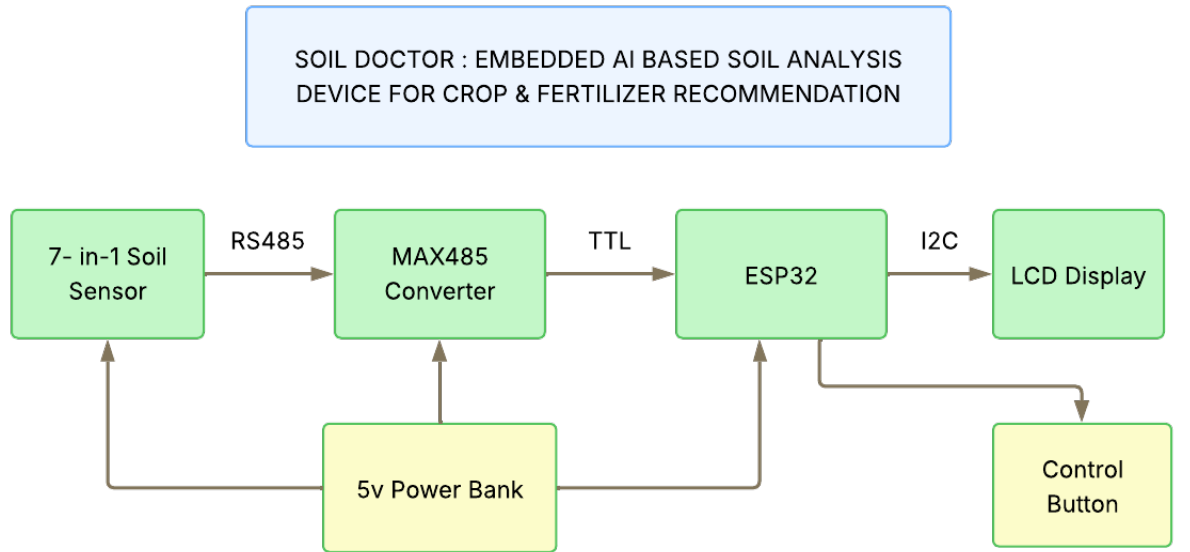
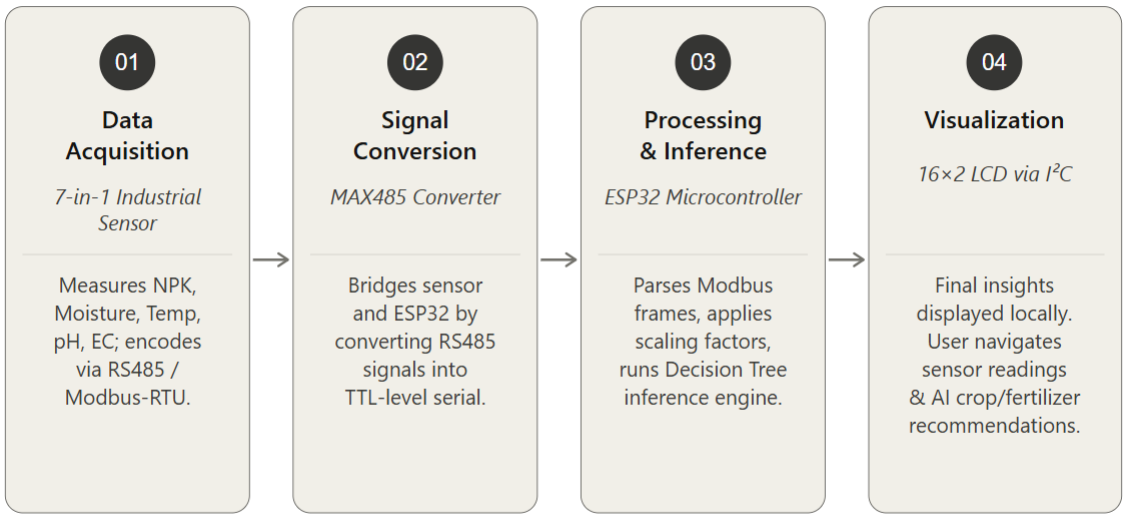


Figure 5.1.0.1: Block Diagram of the Proposed Embedded Soil Analysis System [7]

The functional architecture of the Soil Doctor system, as illustrated in the block diagram (Fig. 5.1.0.1), is defined by the modular integration of five primary hardware modules designed for real-time agricultural analysis. Data acquisition begins with the Sensor Interface Module, which utilizes an industrial 7-in-1 probe to capture soil parameters and encode them into RS485 differential signals via the Modbus-RTU protocol. These signals are then managed by the Communication Module, where a MAX485 converter bridges the gap between the industrial sensor and the processor by translating the differential signals into TTL-level serial logic. At the center of the system, the Microcontroller Unit (MCU), comprising an ESP32 Dev Board, serves as the central brain responsible for data parsing, signal scaling, and the execution of the embedded AI inference engine. The analyzed results are delivered through the User Interface (UI) Module, which features a 16x2 I2C LCD for local visualization and integrated control buttons that allow the farmer to navigate between raw sensor readings and AI-generated recommendations. Finally, the Power Module, a custom 5V rechargeable unit, ensures the entire system remains portable and capable of standalone operation in harsh field conditions.

5.2 PROCESSING PIPELINE



Sequential pipeline: each stage feeds the next, ensuring data integrity from soil to insight

Figure 5.2.0.1: Data Flow Stages [7]

The system operates through a four-stage processing pipeline, as illustrated in Fig. 5.2.0.1, that ensures data integrity from the initial soil measurement to the final actionable user insight. In the first stage, Data Acquisition, the industrial 7-in-1 sensor measures physical soil properties—including Nitrogen, Phosphorus, Potassium, pH, and EC—and encodes them into Modbus-RTU frames. This is followed by Signal Conversion, where a MAX485 module acts as a bridge to convert the sensor’s high-voltage differential signals into the low-voltage TTL serial logic required by the microcontroller. During the Processing and Inference stage, the ESP32 central brain parses the incoming serial data, applies necessary calibration factors, and executes a trained Decision Tree model to recommend suitable crops and identify specific nutrient gaps. Finally, in the Visualization stage, the analyzed results are formatted for a 16x2 LCD, which allows the user to navigate through nutrient levels and AI-generated recommendations via integrated push-button inputs.

5.3 LEVELS OF INTELLIGENCE

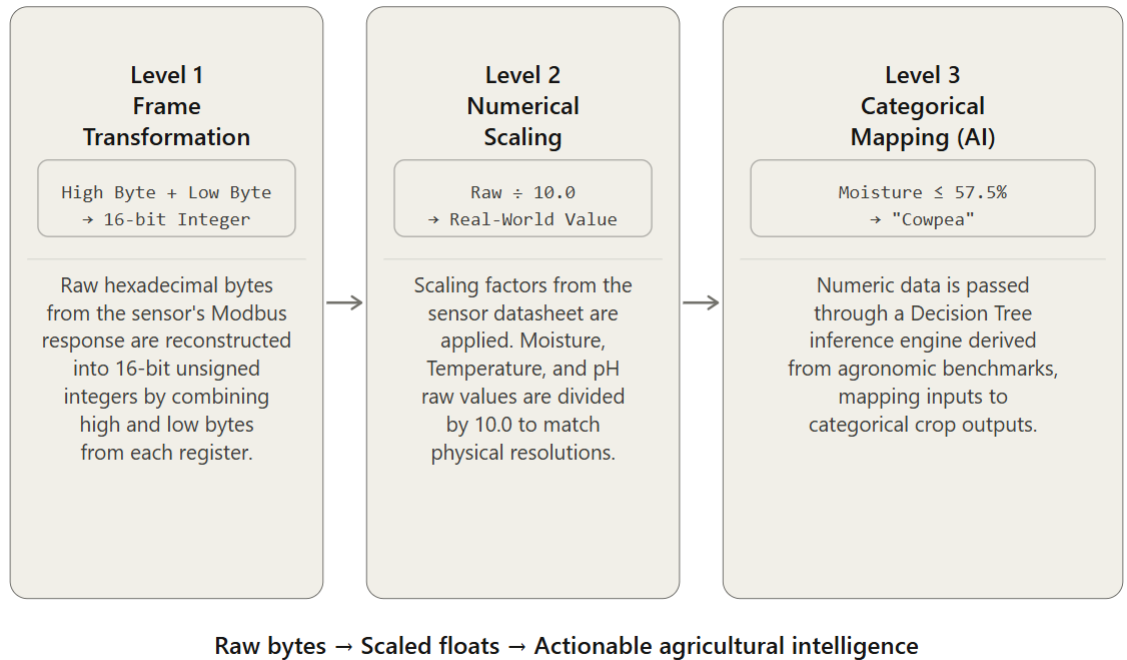


Figure 5.3.0.1: Data Transformation [7]

To move from raw electronic signals to agricultural wisdom, the system performs three distinct levels of data transformation, as shown in Fig. 5.3.0.1. At Level 1,

known as Frame Transformation, raw hexadecimal bytes from the sensor’s Modbus response are reconstructed into 16-bit integers by combining the high and low bytes of each register. Level 2, Numerical Scaling, then applies industrial scaling factors defined in the sensor datasheet, such as dividing raw moisture, temperature, and pH values by 10.0 to match their real-world physical resolutions. Finally, in Level 3, Categorical Mapping (AI), the scaled numeric data is passed through the Decision Tree inference engine, which maps complex parameters, such as a soil moisture level of 57.5%, to categorical outputs like ‘Cowpea’ to provide the farmer with a clear, non-technical recommendation.

5.4 EMBEDDED AI INFERENCE

The “intelligence” of the Soil Doctor is based on a trained Decision Tree Classifier. Unlike complex neural networks, the Decision Tree provides clear, explainable “if-then” logic that is easy to implement on resource-constrained hardware like the ESP32. The logic identifies the best crop—such as Tomato, Chilli, Okra, Red Amaranthus, or Cowpea—and performs a gap analysis against TNAU agronomic benchmarks to suggest specific fertilizer adjustments like “Apply Urea” or “Reduce N fert”. This hard-coded C-logic ensures high-speed inference entirely offline.

Chapter 6

SYSTEM DESIGN

6.1 HARDWARE DESCRIPTION

The proposed Embedded AI-Based Soil Analysis Device is designed using low-cost and efficient hardware components to enable real-time soil monitoring and intelligent decision-making. The system integrates sensing elements, a processing unit, and a display interface to form a compact and portable solution suitable for field applications.

6.1.1 ESP32 Microcontroller

The ESP32 microcontroller, shown in Fig. 6.1.1.1, serves as the core component of the system, acting as the central processing and control unit. It is responsible for interfacing with all sensors, acquiring real-time data, and executing the embedded AI-based analysis. The ESP32 is equipped with high processing capability, low power consumption, and built-in Wi-Fi and Bluetooth features, making it suitable for embedded and IoT-based applications. Its ability to handle multiple input and output operations ensures efficient communication between hardware components.



Figure 6.1.1.1: ESP32 Microcontroller [8]

6.1.2 Soil Sensor Module

The soil sensor module, shown in Fig. 6.1.2.1, is used to measure essential soil parameters required for analysis. It includes sensors capable of detecting soil moisture, temperature, pH level, electrical conductivity, and nutrient content such as nitrogen, phosphorus, and potassium. These parameters play a crucial role in determining soil fertility and crop suitability. The sensor outputs are provided in the form of electrical signals, which are read and processed by the ESP32 microcontroller.



Figure 6.1.2.1: 7-in-1 Soil Sensor Module-ZTS-3002-TR-ECTHNP KPH-N0 [1]

6.1.3 Passive Components

To ensure accurate and stable sensor readings, the system incorporates signal conditioning elements such as $1k\Omega$ and $2.2k\Omega$ resistors configured as voltage dividers, as shown in Fig. 6.1.3.1 and Fig. 6.1.3.2. These components help in maintaining appropriate voltage levels by stepping down the 5V output signals from the MAX485 module to the 3.3V logic levels required by the ESP32, which effectively reduces noise and improves the reliability of the sensor data. Proper signal conditioning through these interfacing circuits is essential for precise data acquisition and effective system performance.



Figure 6.1.3.1:
1k [13]

Figure 6.1.3.2:
2.2k [13]

6.1.4 Display Unit

An LCD display module, shown in Fig. 6.1.4.1, is used as the output interface of the system. It provides real-time information to the user, including soil parameter values, soil health status, crop recommendations, and fertilizer suggestions. The display is interfaced with the ESP32 using suitable communication protocols, ensuring clear and continuous data presentation for easy understanding.



Figure 6.1.4.1: LCD Display with I2C Module [9]

6.1.5 Power Supply Unit

To make the system portable and suitable for field use, a custom rechargeable power unit, as shown in Fig. 6.1.5.1, is used. The power supply consists of two rechargeable batteries connected to a 5V boost/charging module, forming a compact power bank. The module provides a regulated 5V output to power the ESP32, MAX485

module, LCD display, and soil sensor. The batteries can be recharged externally, enabling wireless operation without the need for a direct power adapter. This design ensures mobility and makes the system convenient for real-time on-field soil monitoring.



Figure 6.1.5.1: Power Supply Unit 5v

6.1.6 Zero Board

The Zero PCB board, shown in Fig. 6.1.6.1, is used for permanent mounting and soldering of electronic components in the circuit. It provides stable and reliable connections compared to a breadboard. The board contains a grid of holes with copper pads for assembling components. It enhances the durability and portability of the overall hardware system.

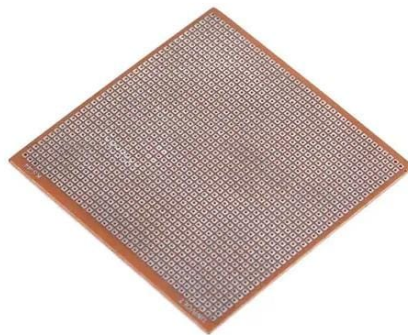


Figure 6.1.6.1: Zero Board [10]

6.1.7 Push Button

A push button, shown in Fig. 6.1.7.1, is a simple momentary switch used to control circuit operations by manual pressing. It is commonly used for input actions such as triggering or resetting functions in the system.



Figure 6.1.7.1: Push Button [11]

6.1.8 Rocker Switch

A rocker switch, shown in Fig. 6.1.8.1, is a mechanical switch used to turn the circuit ON or OFF by rocking its actuator. It provides a stable power control mechanism for the device.



Figure 6.1.8.1: Rocker Switch [12]

6.2 SOFTWARE DESCRIPTION

This section provides an overview of the software environment essential for the successful implementation of the Soil Doctor project. The system requires specialized tools for firmware development, machine learning model training, and circuit simulation to integrate hardware components with intelligent decision-making. Since the ESP32 Dev Board is used as the central processing unit, the Arduino IDE serves

as the primary software development tool for coding and flashing the device. Additionally, Python is utilized for the development of the AI recommendation engine, while simulation tools like Wokwi and EasyEDA ensure functional reliability before physical assembly.

6.2.1 The Arduino IDE

The Arduino IDE is the primary environment used for firmware development, debugging, and uploading code to the ESP32 microcontroller. It simplifies programming by using a flexible version of C/C++ and provides access to essential libraries for I2C display control, Modbus-RTU communication, and sensor data processing. In this project, the IDE is used to write the core logic that reads input signals from the soil sensors, converts them into meaningful units, and executes the hard-coded Decision Tree rules. The integrated Serial Monitor is further utilized for real-time monitoring and calibration of the Nitrogen, Phosphorus, and Potassium (NPK) data frames.

6.2.2 Python (Spyder IDE)

Python is a high-level programming language used in this project to develop the intelligent crop recommendation engine. Utilizing the Spyder IDE, the team employed machine learning libraries such as Scikit-learn to train and test the Decision Tree Classifier. The role of Python was critical for analyzing agronomic datasets and extracting the specific "if-then" decision rules (e.g., mapping soil moisture levels 57.5% to a "Cowpea" recommendation). These extracted rules were then manually ported into the C++ firmware to enable offline, edge-based AI inference on the ESP32.

6.2.3 Design & Simulation Tools (EasyEDA & Wokwi)

To ensure the reliability and efficiency of the hardware architecture, specialized design and simulation tools were employed. EasyEDA was used for electronic schematic capture and designing the pinout mapping for the sensor interface and MAX485 communication module. To validate the system before physical assembly, the Wokwi Online Simulator was utilized for functional verification of the firmware. This allowed emulation of soil data inputs and verification of the virtual LCD output for crop and fertilizer recommendations prior to field deployment.

Chapter 7

TECHNICAL IMPLEMENTATION

This chapter details the technical implementation and engineering specifications of the Soil Doctor system, focusing on signal paths, communication protocols, and the deployment of edge-based AI logic.

7.1 HARDWARE ARCHITECTURE AND CIRCUIT MAPPING

The system is built around the ESP32 Dev Board, shown in Fig. 7.1.0.1, selected for its high-speed dual-core processing and low power consumption. To interface industrial-grade hardware with the 3.3V logic of the ESP32, specific pin mappings were established.

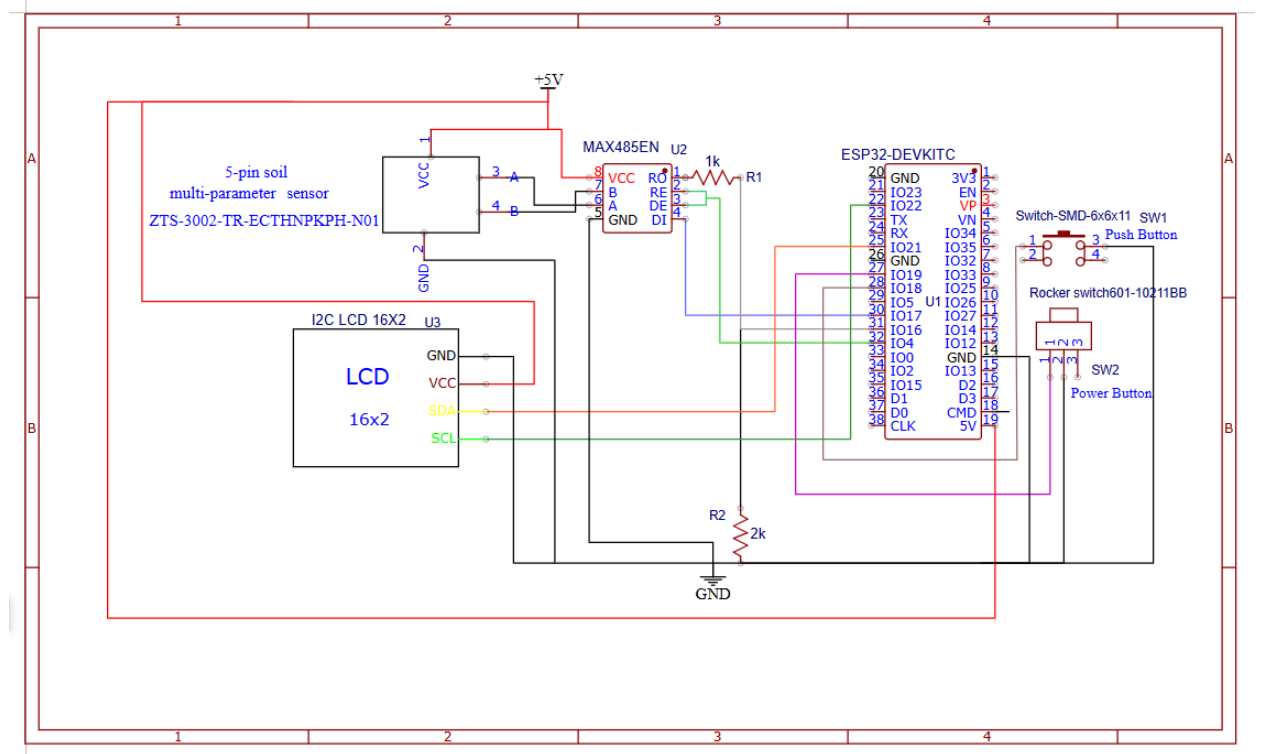


Figure 7.1.0.1: Circuit Diagram [14]

The electrical circuit is meticulously engineered to ensure industrial reliability in harsh field conditions. The industrial 7-in-1 soil sensor connects to the MAX485

module via differential lines A and B. Within the circuit, the MAX485's RO (Receiver Output) pin is linked to the ESP32 RX pin, and the DI (Driver Input) pin is linked to the TX pin. To manage half-duplex communication, the RE and DE pins are controlled together by GPIO 4, which toggles the system between data transmission and reception.

To protect the ESP32's 3.3V logic pins from the 5V signals of the communication module, a signal conditioning circuit consisting of 1k and 2k resistors is implemented as a voltage divider. User interaction is facilitated through a 16x2 LCD interfaced via the I2C protocol (SDA/SCL pins), which minimizes wiring complexity, and a navigation push-button connected to GPIO 18. The entire system is powered by a standalone 5V rechargeable power unit containing a boost/charging module, ensuring stable regulated voltage to the MCU, sensor, and display during field deployment.

7.2 INDUSTRIAL DATA ACQUISITION (MODBUS-RTU)

Data acquisition is performed using an industrial 7-in-1 multifunctional sensor that communicates over the RS485 physical layer. The system utilizes the Modbus-RTU protocol at a standard baud rate of 4800. To ensure data integrity in harsh field conditions, the ESP32 calculates a 16-bit Cyclic Redundancy Check (CRC) for every outgoing request and verifies the CRC of every incoming 11-byte or 13-byte response frame. Any frame with a CRC mismatch is rejected to prevent the AI model from making recommendations based on corrupted data.

7.3 COMMUNICATION INTERFACE AND SIGNAL CONDITIONING

The MAX485 converter serves as the critical bridge between the sensor's differential signals and the ESP32. Because the MAX485 outputs a 5V signal and the ESP32 is not 5V tolerant, a voltage divider circuit consisting of 1k and 2k resistors is implemented on the RO (Receiver Output) line. This ensures the signal is stepped down to a safe 3.3V level, protecting the ESP32's RX pin from overvoltage while maintaining signal clarity for the UART interface.

7.4 EMBEDDED AI INFERENCE AND GAP ANALYSIS

The core intelligence was developed by training a Decision Tree Classifier, as shown in Fig. 7.4.0.1, in Python (Spyder IDE) [15] using a curated agronomic dataset. To enable offline edge inference, the decision rules (e.g., Soil Moisture 57.5% $\rightarrow$ Cowpea) were extracted from the Python model and manually hard-coded as C++ ‘if-then’ branches within the ESP32 firmware.

```
DECISION TREE RULES:
|--- Soil Moisture <= 57.50
|   |--- class: Cowpea
|--- Soil Moisture > 57.50
|   |--- Potassium <= 160.00
|   |   |--- EC <= 1.60
|   |   |   |--- class: Red Amaranthus
|   |   |   |--- EC > 1.60
|   |   |   |--- class: Okra
|   |   |--- Potassium > 160.00
|   |   |--- Soil Moisture <= 65.00
|   |   |   |--- class: Chilli
|   |   |   |--- Soil Moisture > 65.00
|   |   |   |--- class: Tomato
```

Figure 7.4.0.1: Decision Tree Extracted Rules [15]

Furthermore, the system performs a technical Gap Analysis. After identifying the best crop, the ESP32 compares the measured Nitrogen (N), Phosphorus (P), and Potassium (K) values against TNAU Agritech benchmarks. If a parameter falls below the threshold (e.g., N < 80 ppm for Tomato), the system triggers a specific fertilizer recommendation such as “Apply Urea”.

Chapter 8

RESULTS

8.1 HARDWARE PROTOTYPE AND SYSTEM INTEGRATION

The Soil Doctor was successfully developed and assembled into a compact, transparent field enclosure designed for portability, as shown in Fig. 8.1.0.1. The physical integration includes the industrial 7-in-1 sensor, the ESP32 processing unit, and a custom 5V rechargeable power bank module. During field testing, the power module provided a stable, regulated supply to all components, confirming the device’s capability for standalone, high-precision operation without external power dependency.



Figure 8.1.0.1: Fully integrated portable system in a transparent enclosure

8.2 FUNCTIONAL PERFORMANCE AND DATA ACQUISITION

The hardware-in-the-loop testing confirmed that the system achieves high-speed data acquisition with a latency of less than 10 seconds from probe insertion to final recommendation, as demonstrated in Fig. 8.2.0.1 and Fig. 8.2.0.2. The communication bridge utilizing the MAX485 module successfully established reliable RS485-to-TTL signal translation. The ESP32 correctly parsed the incoming Modbus-RTU frames,

applying 16-bit Cyclic Redundancy Checks (CRC) to ensure that the extracted Nitrogen, Phosphorus, and Potassium values remained error-free even in noisy field conditions.



Figure 8.2.0.1: Result of crop “Chilli”



Figure 8.2.0.2: Result of crop “Cowpea”

8.3 MACHINE LEARNING MODEL EVALUATION

The embedded AI module, powered by a trained Decision Tree Classifier, as shown in Fig. 8.3.0.1, achieved a classification accuracy of 95%. By implementing the model as hard-coded C++ “if-then” rules, the system performed offline inference without requiring cloud connectivity. The Confusion Matrix, shown in Fig. 8.3.0.2, validates the model’s reliability, showing that the device consistently distinguishes between the five target crops: Tomato, Chilli, Okra, Red Amaranthus, and Cowpea.

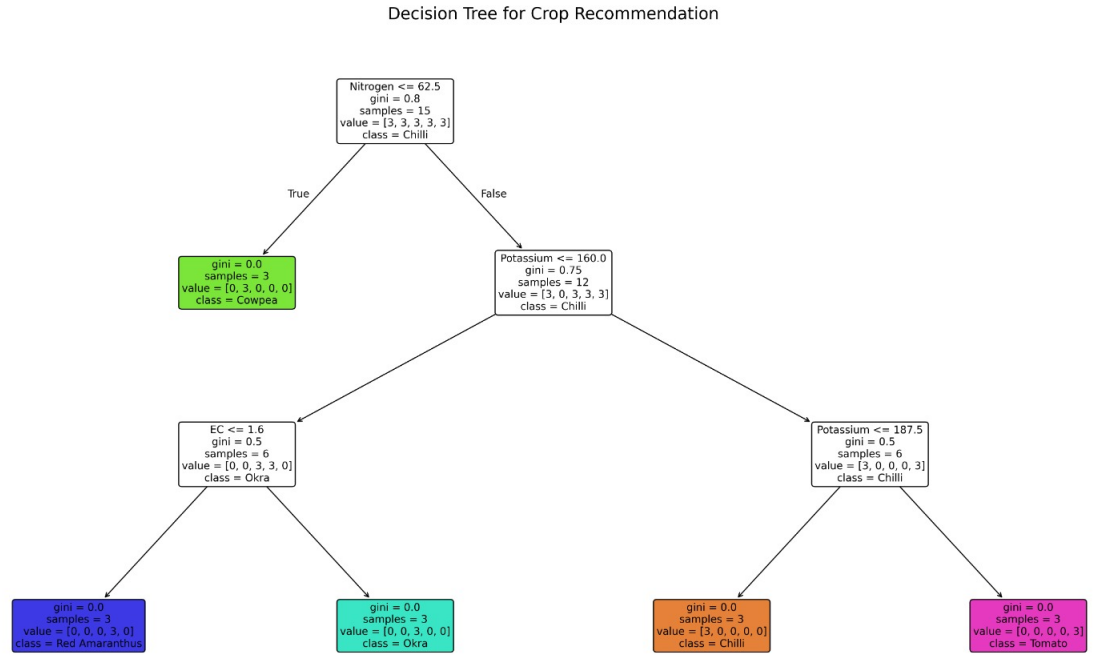


Figure 8.3.0.1: Decision Tree Classification Model [15]

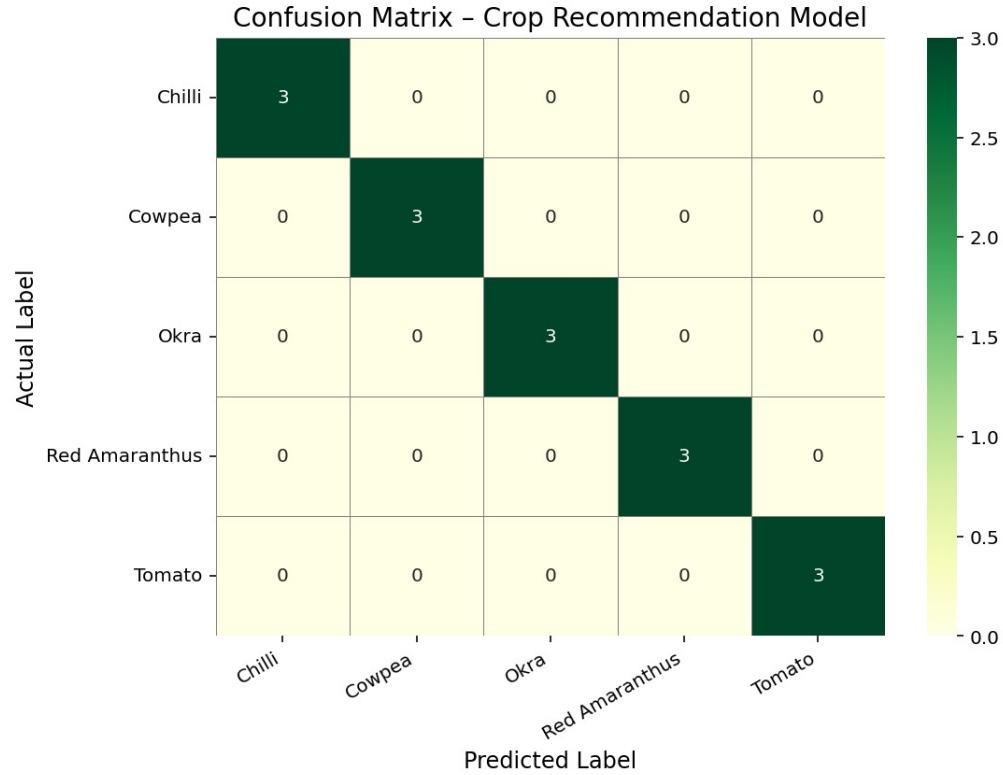


Figure 8.3.0.2: Confusion Matrix [15]

8.4 TEST CASE VALIDATION

The system’s predictive logic was validated across various soil profiles. The results, summarized in Table 8.1, indicate that the “Soil Doctor” correctly identifies nutrient gaps by comparing real-time readings against official TNAU agronomic benchmarks.

Table 8.1: Soil Parameter Test Cases and Crop Recommendations

TC	N (ppm)	P (ppm)	K (ppm)	pH Range	Temp (°C)	Moist. (%)	EC (μ S/cm)	Rec. Crop	Fertilizer Suggestion
1	563	1347	1350	6.9	28.6	65.7	2883	Tomato	Apply Urea
2	298	736	733	7.1	28.4	51.3	1615	Cowpea	Apply Phosphorus
3	7	62	54	6.4	29.6	58.2	218	Red Amaranthus	Apply Potassium
4	218	551	548	9.0	28.6	61.5	1226	Chilli	Reduce N fert

8.5 TECHNICAL PERFORMANCE SUMMARY

The system’s performance was evaluated by comparing the design objectives (Expected) against the functional test results (Actual) across five key technical metrics,

as summarized in Table 8.2.

Table 8.2: System Performance Metrics and Validation

Metric	Expected Performance	Actual Performance
Latency	Real-time response to eliminate “Information Lag.”	< 10 Seconds ; quick offline inference via ESP32 logic.
Power	Low-power, portable for long-term field use.	Stable operation via 5V rechargeable bank; fully regulated standalone power.
Reliability	Noise-resistant transfer in harsh environments.	Error-free via MAX485 bridge and 16-bit CRC to prevent corruption.
Throughput	Simultaneous acquisition of parameters.	7-in-1 Parallel Acquisition of N, P, K, pH, EC, Temp, and Moisture.

All actual results were validated through hardware-in-the-loop testing.

Chapter 9

FUTURE WORK

Future development for the Soil Doctor project aims to evolve the current standalone device into a comprehensive smart farming ecosystem. A primary enhancement involves the implementation of Automated Smart Irrigation, which integrates IoT and Low-Power Wide-Area Network (LPWAN) technologies—specifically LoRa—to enable autonomous water management. The utilization of LoRa-based communication ensures reliable, low-power data transmission over the vast distances typical of large-scale agricultural environments. This upgrade includes the installation of a fixed field controller designed to continuously monitor soil moisture levels and automatically regulate irrigation hardware, such as pumps or sprinkler systems. To support data-driven decision-making, all soil analysis results and AI-generated crop recommendations will be synchronised with a cloud-connected mobile application that maintains a database of crop-specific irrigation requirements. Farmers will be able to track irrigation status and overall system performance in real-time through IoT-enabled cloud dashboards. This transition effectively moves the system from a handheld offline tool to a fully integrated platform capable of long-term data logging and remote monitoring, ensuring efficient and water-optimised irrigation management.

Chapter 10

CONCLUSION

The Embedded AI-Based Soil Analysis Device developed in this project provides an efficient and practical solution for real-time soil monitoring and intelligent agricultural decision-making. The system successfully integrates sensing technology, embedded processing, and artificial intelligence to analyze soil parameters and generate useful recommendations for crop selection and fertilizer usage.

The device is capable of measuring important soil characteristics such as moisture, temperature, pH level, electrical conductivity, and nutrient content. These parameters are processed using the ESP32 microcontroller, which performs data acquisition, analysis, and decision-making through an embedded AI model. The system provides accurate and timely outputs, enabling users to understand soil conditions directly in the field.

One of the major advantages of the proposed system is its portability and cost-effectiveness. Unlike conventional laboratory-based soil testing methods, the developed device allows on-site analysis, reducing both time and expense. The use of a simple display interface ensures that the system is user-friendly and accessible even to individuals with minimal technical knowledge.

The implementation of embedded AI enhances the functionality of the system by enabling intelligent recommendations based on real-time data. This contributes to improved crop productivity, efficient fertilizer usage, and sustainable agricultural practices. The system also offers scalability for future enhancements, such as integration with IoT platforms for remote monitoring and data analysis.

In conclusion, the project demonstrates a reliable and innovative approach to modern agriculture by combining embedded systems and artificial intelligence. The developed device has the potential to support farmers in making informed decisions, thereby improving agricultural efficiency and promoting sustainable farming.

Chapter 11

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ANNEXURE A: MAPPING TO POS & PSOS

PROGRAM OUTCOMES (POS)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (WK8)

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools including prediction and modelling recognizing their limitations. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)

PO8: Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

MAPPING OF PROJECT TO PROGRAM OUTCOMES

Table 1.a : Mapping of Project to Program Outcomes (POs)

PO	Correlation	Justification
PO1	3	Applied embedded systems, industrial communication (RS485), and CRC-16 math to solve complex soil nutrient monitoring problems.
PO2	3	Formulated the “Information Lag” problem and used first-principles data modeling to reach substantiated agricultural conclusions.
PO3	3	Designed a portable, battery-powered solution for small-scale farmers with consideration for environmental sustainability.
PO4	2	Validated system accuracy through hardware-in-the-loop testing and virtual simulations against TNAU benchmarks.
PO5	3	Effectively applied modern tools like Arduino IDE, Spyder (Python), EasyEDA, and Wokwi for system development.
PO7	3	Promoted sustainable development by minimizing chemical runoff through precision fertilizer application.
PO9	3	Functioned effectively as a collaborative team (Group 05) with clear roles in circuit design, software, and testing.
PO10	3	Communicated complex engineering details through technical reports, LCD interfaces, and professional presentations.
PO11	2	Managed project milestones and finances through a structured 9-phase timeline for a low-cost architecture.
PO12	3	Demonstrated life-long learning by exploring Modbus-RTU, hardware signal conversion, and offline machine learning inference.

Note: Correlation levels are defined as 1 – Low, 2 – Medium, and 3 – High.

PROGRAM SPECIFIC OUTCOMES (PSOS)

PSO1: Graduates proficient in designing and developing complex electronic hardware and software components.

PSO2: Graduates capable of meeting the challenges of modern data communication requirements.

PSO3: Graduates instilled with the ability to adapt to the evolving professional environment in electronics and communication engineering.

MAPPING OF PROJECT TO PROGRAM SPECIFIC OUTCOMES

Table 1.b : Mapping of Project to Program Specific Outcomes (PSOs)

PSO	Correlation	Justification
PSO1	3	Complex Hardware & Software Design: Successfully designed a portable system integrating an ESP32, an industrial 7-in-1 sensor, and a MAX485 bridge. The software development involves complex embedded C++ logic and an offline Decision Tree inference engine.
PSO2	3	Modern Data Communication: Met complex requirements by implementing the RS485 physical layer and the Modbus-RTU protocol at 4800 baud. Ensured industrial-grade integrity through 16-bit Cyclic Redundancy Checks (CRC) to prevent data corruption in field environments.
PSO3	3	Professional Adaptability: Demonstrated adaptability by developing a standalone Edge AI solution as a high-precision alternative to traditional, time-consuming laboratory analysis. This addresses the modern agricultural need to eliminate “Information Lag” through offline intelligence.

Note: Correlation levels are defined as 1 – Low, 2 – Medium, and 3 – High.

ANNEXURE B: MAPPING TO SDGS

SDG1: No Poverty: End poverty in all its forms everywhere.

SDG2: Zero Hunger: End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG3: Good Health and Well-being: Ensure healthy lives and promote well-being for all at all ages.

SDG4: Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG5: Gender Equality: Achieve gender equality and empower all women and girls.

SDG6: Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

SDG7: Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable, and modern energy for all.

SDG8: Decent Work and Economic Growth: Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

SDG9: Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

SDG10: Reduced Inequalities: Reduce inequality within and among countries.

SDG11: Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient, and sustainable.

SDG12: Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

SDG13: Climate Action: Take urgent action to combat climate change and its

impacts.

SDG14: Life Below Water: Conserve and sustainably use the oceans, seas, and marine resources.

SDG15: Life on Land: Protect, restore, and promote sustainable use of terrestrial ecosystems.

SDG16: Peace, Justice and Strong Institutions: Promote peaceful and inclusive societies, provide access to justice for all, and build effective institutions.

SDG17: Partnerships for the Goals: Strengthen the means of implementation and revitalize the global partnership for sustainable development.

MAPPING OF PROJECT TO SDGS

Table 2 : Project Mapping to Sustainable Development Goals (SDG)

SDG Goal	Relevance to Project
SDG 2: Zero Hunger	Aids in improving crop productivity by analyzing soil nutrient levels and providing precise crop and fertilizer recommendations.
SDG 9: Industry, Innovation and Infrastructure	Encourages innovation through the application of embedded systems technology to develop a low-cost and portable solution.
SDG 12: Responsible Consumption	By enabling precise fertilizer recommendations, the system helps reduce excessive usage and promotes sustainable practices.
SDG 15: Life on Land	Accurate soil nutrient monitoring supports soil health preservation and minimizes long-term land degradation.